

Lion Guardians Patrols Effort

Technical Guide

Patrol Effort Analysis — Methodology & Calculation Reference
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1. Overview

The **LG Patrols Effort** workflow analyses ranger patrol activity in the Amboseli ecosystem for the **Lion Guardians** programme. It ingests patrol tracks and patrol events from **EarthRanger**, computes patrol effort maps and summary statistics per guardian and patrol type, and delivers an interactive dashboard plus a print-ready Word report.

The workflow produces three maps per group (Patrol Events, Patrol Trajectories, Linear Time Density), five scalar metrics, two charts (bar chart and pie chart), and six summary CSV tables.

Note: All per-group outputs are produced by iterating over a user-chosen grouper: patrol type, patrol serial number, or patrol subject.

2. Dependencies & Prerequisites

2.1 EarthRanger Connection

All patrol data is fetched from an **EarthRanger** instance via `set_er_connection`. The workflow uses `set_patrols_and_patrol_events_params` to configure the query, then calls `get_patrols_from_combined_params` and `get_patrol_observations_from_patrols_df_and_combined_params` to retrieve patrol tracks, and `unpack_events_from_patrols_df_and_combined_params` for events. The sub-page size is 200 records; patrols are not filtered to overlap the date range only.

2.2 Groupers

Three grouper fields are available via `set_groupers`:

Grouper field	EarthRanger source	Typical use
patrol_type	patrol.patrol_type.value	One output per patrol type
patrol_serial_number	patrol.serial_number	One output per patrol serial
patrol_subject	patrol.leader / members	One output per guardian ranger

2.3 Static Geodata Files

Three boundary datasets are downloaded from Dropbox and cached locally:

Dataset	File	Purpose
Group Ranch Boundaries	lg_group_ranch_boundaries.gpkg	Community ranch polygons in Amboseli
Conflict Hotspot Areas	lg_conflict_hotspots.gpkg	Known human–lion conflict hotspot features
Protected Areas	lg_protected_areas.gpkg	National parks and reserves

All three files use `overwrite_existing: false` (3 retries). After loading, each is reprojected to **EPSG:4326** and annotated with its geometry type before layer creation.

2.4 Word Document Templates

Template file	Purpose
patrol_guardians_cover_page.docx	Report cover page — period, preparer

Template file	Purpose
custom_patrol_template.docx	Per-grouper section — maps, charts, and summary tables

2.5 Base Map Tile Layers

Layer	Opacity	Max zoom
ArcGIS World Hillshade	100 %	20
ArcGIS World Street Map	15 %	20

The hillshade provides full-opacity terrain context. The street map is overlaid at 15 % to show roads and settlement names without obscuring terrain or patrol data.

3. Data Ingestion Pipeline

3.1 Patrol Observations → Relocations → Trajectories

`process_relocations` converts raw patrol observations to a standardised `GeoDataFrame`. Retained columns include patrol identifiers (`patrol_id`, `patrol_start_time`, `patrol_end_time`, `patrol_serial_number`, `patrol_subject`), fix quality (`junk_status`), and point geometry. Three null-island coordinate pairs are filtered out:

- (180.0, 90.0) — boundary sentinel
- (0.0, 0.0) — null-island artefact
- (1.0, 1.0) — common default / test value

`relocations_to_trajectory` then connects consecutive fixes per patrol into `LineString` segments, adding `dist_meters`, `speed_kmh`, `segment_start`, and `segment_end`. Trajectories are persisted as `trajectories.geoparquet`.

3.2 Patrol Events

`unpack_events_from_patrols_df_and_combined_params` extracts associated events for each patrol. `get_event_type_display_names_from_events` enriches the events `GeoDataFrame` with human-readable event type names (`append_category_names`: `duplicates`). Events are converted to the user timezone and persisted as `events.geoparquet`.

3.3 Temporal Index & Column Renaming

`add_temporal_index` keys the trajectory `GeoDataFrame` to `extra_patrol_start_time`, grouped by the configured groupers, enabling per-group iteration. Four columns are then renamed via `map_columns` (`raise_if_not_found`: `true`):

Original column	Renamed to
<code>extra_patrol_type_value</code>	<code>patrol_type</code>
<code>extra_patrol_serial_number</code>	<code>patrol_serial_number</code>
<code>extra_patrol_status</code>	<code>patrol_status</code>
<code>extra_patrol_subject</code>	<code>patrol_subject</code>

The renamed `GeoDataFrame` is split into per-group partitions by `split_groups`. All downstream map and metric tasks iterate over these partitions via `mapvalues`.

3.4 Trajectory Colormap

A user-selectable **Trajectory Category** field determines how patrol tracks are coloured. Available options are `patrol_type`, `patrol_subject`, and `patrol_serial_number`. `apply_color_map` maps the chosen column to the **Paired** colormap, writing the result to `patrol_traj_colormap`. Similarly, patrol events are coloured by `event_type` using the **Accent** colormap (`event_type_colormap`).

4. Static Map Layers

Four static layers are built once and composited onto every group-level map to provide spatial context.

4.1 Layer Styles

Layer	Colour (RGB)	Opacity	Filled	Notes
Group Ranch Boundaries	(169, 169, 169) grey	55 %	No	Outline only, line width 4.5
Conflict Hotspots	(220, 20, 60) crimson	75 %	Yes	Point radius 2.55, line width 1.95
Protected Areas	(77, 102, 0) dark green	35 %	Yes	Line width 1.95
Hotspot Text Labels	(20, 20, 20) near-black	—	—	Arial, 1 000 m base, 40–75 px clamp, centroid-anchored

5. Map Outputs — Methodology

5.1 Patrol Events Map

`create_scatterplot_layer` renders each patrol event as a point marker, coloured by `event_type_colormap` (Accent colormap). Point radius is 5 px at 75 % opacity with stroked outlines. Before rendering, geometric outliers are removed via `exclude_geom_outliers` (z-threshold: 3) and null geometries are dropped. The event layer is combined with the four static boundary layers. The map is auto-zoomed to the event extent (expansion factor 1.05) and persisted as HTML (suffix: `events`), then converted to PNG at 2× scale with a 40 s tile-load wait.

5.2 Patrol Trajectories Map

`create_path_layer` renders patrol track segments coloured by the user-selected Trajectory Category column:

Property	Value
Colour	<code>patrol_traj_colormap</code> (Paired colormap, per-category)
Width	1.85 px, min 2 px, max 6 px (screen-space pixels)
Cap / Join	Rounded
Opacity	55 %

The path layer is combined with the four static layers and auto-zoomed to the trajectory extent (expansion factor 1.05). The map is persisted as HTML (suffix: `patrol_trajectories`). PNG screenshot generation is available but disabled by default in the current workflow version.

5.3 Linear Time Density Map

`create_meshgrid` builds a raster grid over the full patrol trajectory extent. `calculate_linear_time_density` then computes the time-weighted density of patrol coverage. Parameters:

Parameter	Value	Meaning
percentiles	50, 60, 70, 80, 90, 100	Contour probability thresholds extracted as polygons
intersecting_only	false	Meshgrid covers the full AOI extent

NaN-percentile rows are dropped and the result is sorted ascending. Contour polygons are coloured with the **RdYlGn** diverging colormap (innermost 50th percentile = red, outermost 100th = green) at 45 % opacity. The map is persisted as HTML (suffix: `time_density`) and converted to PNG at 2× scale with a 40 s tile-load wait.

6. Summary Metrics

6.1 Per-Guardian Statistics

`summarize_df` aggregates trajectory data grouped by `patrol_subject`:

Output column	Source column	Aggregator	Output unit
<code>no_of_patrols</code>	<code>extra__patrol_id</code>	<code>nunique</code>	count
<code>total_distance</code>	<code>dist_meters</code>	<code>sum</code>	km
<code>total_time</code>	<code>timespan_seconds</code>	<code>sum</code>	h

A separate summary counts the number of events per guardian (id `nunique`). Results are persisted as CSVs. A pivot table is also generated — events pivoted by `event_type_display` — to show each guardian's event breakdown.

6.2 Patrol Type Summary

The same three aggregations (`no_of_patrols`, `total_distance`, `total_time`) are computed grouped by `patrol_type`. This shows comparative effort across different patrol activity categories.

6.3 Event Type Summary

Events are summarised by `event_type_display` using `id nunique`, giving a count of distinct events per type across the analysis period.

6.4 Monthly Summary

`extract_date_parts` extracts `month`, `day`, and `month_name` from `extra__patrol_start_time`. Patrol effort is then summarised by `month_name` (`no_of_patrols`, `total_distance`, `total_time`) to reveal seasonal patrol patterns.

6.5 Scalar Dashboard Widgets

Five scalar metrics are computed per group from the trajectory data and displayed as single-value widgets (1 decimal place):

Widget title	Source column	Aggregator	Unit
Total Patrols	<code>extra__patrol_id</code>	<code>nunique</code>	count
Total Time	<code>timespan_seconds</code>	<code>sum</code>	h

Widget title	Source column	Aggregator	Unit
Total Distance	dist_meters	sum	km
Average Speed	speed_kmhr	mean	km/h
Max Speed	speed_kmhr	max	km/h

7. Event Charts

7.1 Time Series Bar Chart

`draw_time_series_bar_chart` plots patrol event counts over time. X-axis: time (event timestamp); Y-axis: event_type_display; aggregation: count; category colour: event_type_colormap (Accent colormap). The chart is persisted as HTML (suffix: patrol_events_time_series_bar_chart) and converted to PNG.

7.2 Pie Chart

`draw_pie_chart` shows the proportional breakdown of events by event_type_display, using event_type_colormap for slice colours and displaying raw counts as text labels. The chart is persisted as HTML (suffix: patrols_pie_chart) and converted to PNG.

8. Word Report (.docx)

8.1 Cover Page

`create_guardians_ctx_cover` builds the cover context (report period, *Ecoscope* as preparer). `create_context_page_lg` renders it into context_page.docx using the patrol_guardians_cover_page.docx template.

8.2 Per-Grouper Sections

`guardians_ctx` assembles a context dict per group containing maps (events map, trajectories map, time density map), charts (pie chart, bar chart), and CSV data (monthly efforts, guardian event pivot, guardian events, guardian patrol stats, event type efforts, trajectory DataFrame). `generate_guardians_report` renders each section from the custom_patrol_template.docx template. Image boxes: **3.85 × 6.48 inches**. `validate_images: true` catches missing PNGs before rendering.

8.3 Document Merge

`merge_cl_files` concatenates the cover page and all per-grouper sections into a single Word file saved to the results directory.

9. Interactive Dashboard

`gather_dashboard` assembles the patrol dashboard from ten widget groups:

Widget	Type	Source task
Total Distance	Single value	total_patrol_dist → patrol_dist_grouped_widget

Widget	Type	Source task
Average Speed	Single value	avg_speed → avg_speed_grouped_widget
Max Speed	Single value	max_speed → max_speed_grouped_widget
Total Patrols	Single value	total_patrols → total_patrols_grouped_sv_widget
Total Time	Single value	total_patrol_time → patrol_time_grouped_widget
Patrol Events Map	Map	draw_events → events_grouped_map_widget
Trajectories Map	Map	trajs_ecomap → trajs_grouped_map_widget
Events Bar Chart	Plot	patrol_events_bar_chart → grouped_bar_plot_widget_merge
Events Pie Chart	Plot	patrol_events_pie_chart → patrol_events_pie_widget_grouped
Time Density Map	Map	td_ecomap → td_grouped_map_widget

Note: Widget tasks use skipif: [never] so the dashboard always assembles, even when some groups have no data.

10. Output Files

All files are written to \$ECOSCOPE_WORKFLOWS_RESULTS.

File / pattern	Format	Content
events.geoparquet	GeoParquet	Patrol events with display names and timezone-converted timestamps
trajectories.geoparquet	GeoParquet	Patrol segments with speed_kmh, dist_meters
_events.html	HTML	Interactive patrol events map
_patrol_trajectories.html	HTML	Interactive patrol trajectories map
_time_density.html	HTML	Interactive linear time density map
_events.png	PNG	2× screenshot of patrol events map
_time_density.png	PNG	2× screenshot of time density map
_patrols_pie_chart.png	PNG	2× screenshot of event type pie chart
_bar_chart.png	PNG	2× screenshot of events time series bar chart
_guardian_patrol.csv	CSV	Per-guardian patrol effort (patrols, distance, time)
_guardian_events.csv	CSV	Per-guardian event count
_pivot_guardian_events.csv	CSV	Guardian × event type pivot table
_patrol_types.csv	CSV	Per-patrol-type effort summary
_event_types.csv	CSV	Per-event-type count
_monthly_patrol_efforts.csv	CSV	Monthly patrol effort summary
context_page.docx	Word	Rendered report cover page

File / pattern	Format	Content
.docx	Word	Per-grouper report section
.docx	Word	Final combined Word report

11. Workflow Execution Logic

11.1 Skip Conditions

Two default skip conditions apply to every task (task-instance-defaults):

- **any_is_empty_df** — skips the task (and all dependants) when any input DataFrame is empty, handling patrol types or periods with no data gracefully.
- **any_dependency_skipped** — propagates skips downstream automatically.

Widget and map-widget tasks override this with skipif: [never] to ensure the dashboard always assembles.

11.2 Data Flow Summary

Stage	Tasks
Setup	ER connection, time range, timezone, groupers, base maps
Geodata download	3 boundary files + 2 Word templates from Dropbox
Static layers	Ranch, hotspot, protected area, hotspot text layers
Patrol ingest	Params → prefetch → observations → events → rename → convert TZ
Trajectories	Relocations → trajectories → temporal index → rename → split groups
Events branch	Filter → temporal index → colormap → rename → outlier removal → scatter layer → map → HTML → PNG → widget
Trajectories branch	Speed format → rename → colormap → path layer → map → HTML → widget
Time density branch	Meshgrid → LTD → drop NaN → sort → colormap → GeoJSON layer → map → HTML → PNG → widget
Metrics branch	5 scalar widgets; 6 CSV summaries (guardian, patrol type, event type, monthly, pivot)
Charts branch	Time series bar chart + pie chart → HTML → PNG → widgets
Report assembly	Cover page + per-group sections → merge docx
Dashboard	gather_dashboard combines all 10 widgets

12. Software Versions

Package	Version	Role
ecoscope-workflows-core	0.22.17.*	Core task library and workflow engine
ecoscope-workflows-ext-ecoscope	0.22.17.*	Spatial analysis tasks (LTD, relocations, trajectories)
ecoscope-workflows-ext-custom	0.0.40.*	Utility tasks (column mapping, screenshots, pivots)

Package	Version	Role
ecoscope-workflows-ext-ste	0.0.18.*	Summary table and aggregation tasks
ecoscope-workflows-ext-mnc	0.0.7.*	Pivot table task (pivot_df)
ecoscope-workflows-ext-icf	0.0.0.*	ICF domain tasks
ecoscope-workflows-ext-big-life	0.0.8.*	Big Life Foundation domain tasks
ecoscope-workflows-ext-lion-guardians	0.0.6.*	Lion Guardians Word report rendering tasks

Packages are distributed via the prefix.dev conda channel and pinned to patch-compatible versions (*.*) suffix). The runtime environment is managed by **pixi**.